

Linear Algebra

1. Compute $\begin{pmatrix} 3 \\ 3 \end{pmatrix} - \begin{pmatrix} -8 \\ 7 \end{pmatrix}$.
2. Find the dot product $\begin{pmatrix} -1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 5 \\ 7 \end{pmatrix}$.
3. Find the magnitude of $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.
4. Find a unit vector in the direction of $\begin{pmatrix} -3 \\ -4 \end{pmatrix}$.
5. Compute $\begin{bmatrix} -5 & -4 \\ -1 & 1 \end{bmatrix} - \begin{bmatrix} 6 & 0 \\ -7 & -4 \end{bmatrix}$.
6. Compute $-3 \begin{bmatrix} 3 & 3 \\ 0 & 4 \end{bmatrix}$.
7. Compute the product $\begin{bmatrix} 2 & 4 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ -4 & -4 \\ 1 & -3 \end{bmatrix}$.
8. Find the determinant of $\begin{vmatrix} 4 & 2 \\ -2 & 6 \end{vmatrix}$.
9. Find the inverse of $\begin{bmatrix} -4 & -4 \\ 0 & -1 \end{bmatrix}$.
10. Solve the system represented by the augmented matrix $\left[\begin{array}{cc|c} -3 & -2 & 14 \\ 0 & -2 & 2 \end{array} \right]$.
11. Reduce to reduced row-echelon form: $\begin{bmatrix} -2 & 2 & -14 \\ 0 & -1 & 3 \end{bmatrix}$.
12. Find the rank of $\begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & -3 \\ 0 & 1 & -3 \end{bmatrix}$.
13. Are the columns of $\begin{bmatrix} 8 & -4 \\ -2 & 7 \end{bmatrix}$ linearly independent?
14. Find the coordinate vector of $\begin{pmatrix} -14 \\ -12 \end{pmatrix}$ in the basis formed by the columns of $\begin{bmatrix} -2 & 2 \\ 0 & 3 \end{bmatrix}$.
15. Find the eigenvalues of $\begin{bmatrix} -3 & -3 \\ 0 & 2 \end{bmatrix}$.
16. Find an eigenvector of $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ for $\lambda = -1$.
17. Project $\begin{pmatrix} -6 \\ 6 \end{pmatrix}$ onto $\begin{pmatrix} -4 \\ 1 \end{pmatrix}$.
18. For $T(x) = Ax$ with $A = \begin{bmatrix} 3 & -4 \\ 2 & -4 \end{bmatrix}$, find $T(v)$ when $v = \begin{pmatrix} -2 \\ -2 \end{pmatrix}$.

Solutions

1. Combine corresponding entries: $\begin{pmatrix} 11 \\ -4 \end{pmatrix}$.
2. $(-5) + 21 = 16$.
3. $\|v\| = \sqrt{2} = \sqrt{2}$.
4. Divide by 5: $\begin{pmatrix} -\frac{3}{5} \\ -\frac{4}{5} \end{pmatrix}$.
5. Operate entry by entry: $\begin{bmatrix} -11 & -4 \\ 6 & 5 \end{bmatrix}$.
6. Multiply every entry: $\begin{bmatrix} -9 & -9 \\ 0 & -12 \end{bmatrix}$.
7. Row-by-column multiplication gives $\begin{bmatrix} -13 & -5 \\ 6 & -2 \end{bmatrix}$.
8. $\det(A) = 28$.
9. $A^{-1} = \begin{bmatrix} -\frac{1}{4} & 1 \\ 0 & -1 \end{bmatrix}$.
10. The solution vector is $\begin{pmatrix} -4 \\ -1 \end{pmatrix}$.
11. The RREF is $\begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & -3 \end{bmatrix}$.
12. Its RREF is $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix}$, so $\text{rank}(A) = 2$.
13. $\det(A) = 48$, so the columns are independent.
14. The coordinate vector is $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$.
15. Because the matrix is triangular, $\lambda_1 = -3$, $\lambda_2 = 2$.
16. One valid eigenvector is $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$.
17. $\text{proj}_u(v) = \frac{v \cdot u}{u \cdot u} u = \frac{30}{17} u = \begin{pmatrix} -\frac{120}{17} \\ \frac{30}{17} \end{pmatrix}$.
18. $T(v) = Av = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$.